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A grinding contour prediction model and multi-objective process parameter optimization using functional principal component analysis

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HIGHLIGHTS

- An FPCA-based method for grinding groove contour reconstruction is proposed, surpassing scalar metric limitations.
- A novel Bayesian-optimized ensemble learning model is proposed for accurate small-sample contour prediction.
- Model reduces RMSE by 14.90%, increases R^2 by 8.65%, and lowers MAE by 6.17% over the traditional response surface method.
- Multi-pass grinding is simulated, demonstrating the necessity of dynamic path adjustment based on corresponding contour.

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ABSTRACT

In additive repair manufacturing, the effectiveness of defect removal and the bonding quality at the interface in subsequent additive repairs are influenced by surface topography, which is regulated by robotic grinding parameters. Conventional scalar metrics such as grinding depth or width are insufficient to fully characterize the three-dimensional morphology of grinding grooves. This paper proposes a grinding groove contour reconstruction method based on functional data analysis (FDA) and establishes a contour prediction model that integrates Bayesian optimization and ensemble learning. The contour reconstruction approach employed functional principal component analysis (FPCA) to reduce the dimensionality of high-dimensional contour data. The first three principal components accounted for 97.2 % of the cumulative variance, with reconstruction accuracy reaching RMSE = 0.2364 mm and R^2 = 0.8826. On this basis, the prediction model incorporated Bayesian optimization and ensemble learning, improving contour prediction performance to RMSE = 0.2205 ± 0.0228 mm, R^2 = 0.8420 ± 0.0300, and MAE = 0.1688 ± 0.0206 mm. Compared to a baseline response surface model, the proposed method reduced RMSE by 14.90 %, increased R^2 by 8.65 %, and decreased MAE by 6.17 %, demonstrating significantly enhanced modeling capability for small-sample datasets. Finally, the single-pass grinding contour model was applied to simulate multi-pass material removal, identifying a medium overlap ratio (0.5) as optimal for balancing surface quality and material removal rate. Simulation results indicated that path parameters should be dynamically adjusted according to real-time grinding contours during actual machining. This study advances traditional process optimization from scalar-based metrics to holistic contour prediction and control, providing a useful reference for robotic multi-pass grinding path planning.

1. Introduction

With the development of key industrial sectors such as aerospace, automotive, and energy, robotic intelligent manufacturing has become increasingly prevalent. Its high flexibility, strong adaptability, and ease of integration enable it to effectively meet the machining requirements of complex surfaces and irregularly shaped components [1–4]. In processes such as casting and welding, surface defects such as porosity and weld cracks may occur, as illustrated in Fig. 1-1. These defects must be removed by grinding or grooving to allow subsequent weld repair, which ensures dimensional precision and provides a suitable substrate for additive remanufacturing [5–7]. For example, additive

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manufacturing can deposit material onto existing components, enabling cost-effective repair and reducing the need for part replacement [8,9]. Grinding to create specific grooves is a crucial pre-processing step. The complete contour morphology of these grooves not only governs the effectiveness of defect removal but also directly affects stress distribution and service performance at the bonding interface [10]. Therefore, targeted grinding studies are essential to achieve precise contour control and desirable surface topography. Intelligent modeling and sustainable grinding technologies are increasingly driving gains in precision and efficiency in grinding operations [11].

In robotic grinding process optimization, conventional methods typically employ response surface methodology (RSM) [12,13], neural networks [14], and intelligent algorithm-based parameter optimization models [15,16] to regulate and optimize parameters such as feed rate, normal force, and grinding wheel speed. These methods aim to establish mapping relationships between process parameters and simple geometric characteristics, such as grinding depth and width [17,18]. For example, Yang et al. [19] developed a hybrid multi-task learning (H-MTL) framework for response surface modeling, which respectively achieved high modeling accuracy without additional experimental cost. Xiao et al. [20] combined Optimum Latin Hypercube Design, a surface integrity model, and a Genetic Algorithm-Backpropagation Neural Network (GA-BPNN) to predict grinding parameters for target surface integrity. Their model reported a mean relative error below 7.5 %, with GA-BPNN regression correlations exceeding 0.8 and a maximum prediction error under 6.5 %. Pan et al. [21] developed a MAML-PPOBE meta-reinforcement learning algorithm enabling fast adaptation across grinding tasks with high precision. Kishore et al. [22] implemented a machine-learning-based approach to establish correlations between grinding parameters and outputs, effectively modeling the complex grinding process of Inconel 625. However, these methods suffer from two key limitations: complex parameter coupling often causes poor generalization with limited data, and conventional scalar outputs fail to capture complete contour morphology. Although improvements have been attempted, these fundamental issues persist. For instance, Zhang et al. [23] used RSM to build a structural damage prediction model and optimize grinding parameters for predicting weld bead grinding area. Nevertheless, relying solely on geometric scalars like depth, width, or area cannot accurately capture cross-sectional groove shape or extract advanced morphological features such as asymmetry [24]. In repair pretreatment applications, the full contour morphology is more critical than individual geometric parameters, as it directly affects bonding quality and stress distribution in subsequent additive processes [25]. Inverse optimization deriving optimal inputs from target outputs like depth or contour is crucial for intelligent grinding. Yet current methods remain limited to simple outputs and lack inverse control of complex contour topography, particularly hindering applications in 3D free-form surfaces common in additive repair contexts.

Grinding path planning critically influences both machining efficiency and the consistency of surface quality. Conventional path planning methods, such as the equal pitch and equal residual height approaches, predominantly rely on geometric models or empirical rules [26]. These strategies, however, often overlook the physical interactions of the actual grinding process, resulting in difficulties in maintaining uniform surface quality [27,28]. Recent advances in sensing technology have enabled new approaches that incorporate real-time inspection data into path planning [29]. For example, several studies use 3D laser scanning to capture workpiece point cloud data, adapting path intervals according to detected contour characteristics [30–32]. Cheng et al. [33] introduced a grinding compensation method using a laser vision sensor, combining parametric and point cloud-based trajectory planning with edge-based adjustment strategies. Their work optimized paths by developing mathematical models for material removal and surface contour prediction. Despite these innovations, most current path planning techniques remain heavily dependent on geometric models and lack tight integration with physical process models. This gap often leads to discrepancies between planned paths and actual grinding outcomes [34]. The challenge is particularly evident in multi-pass grinding, where the absence of accurate models for predicting and controlling overlap effects between consecutive passes undermines the consistency of the final surface [35,36].

To address the above problems, this paper proposes a functional principal component analysis (FPCA) contour reconstruction framework and then builds up a grinding contour prediction model, as illustrated in Fig. 1-2. Within this contour reconstruction framework, the entire cross-sectional contour point cloud is processed and reduced via FPCA into a small set of principal component scores. By integrating these within a conventional response surface modeling approach, the method predicts complex grinding contour shapes directly from process parameters. The model achieves an overall RMSE of 0.2591 $\pm$ 0.0176 mm, R^2 of 0.7750 $\pm$ 0.0288, and MAE of 0.1799 $\pm$ 0.0091 mm. To further improve predictive accuracy under small-sample constraints, an enhanced contour prediction model incorporating Bayesian optimization and ensemble learning is introduced. Through FPCA-based feature extraction,

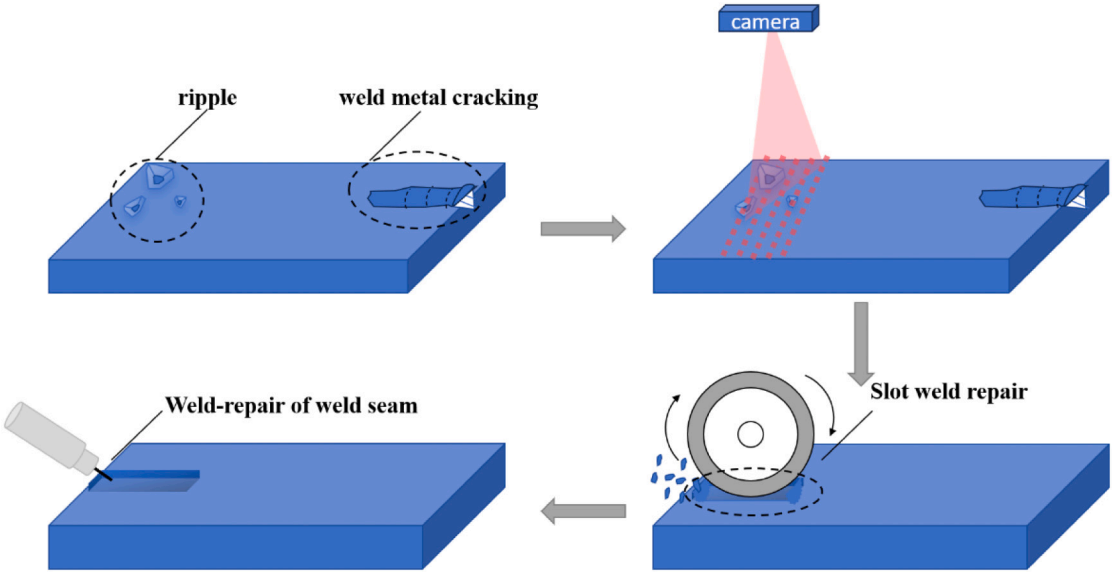


Fig. 1-1. Schematic diagram of defect identification and grinding removal process for castings or welded components.

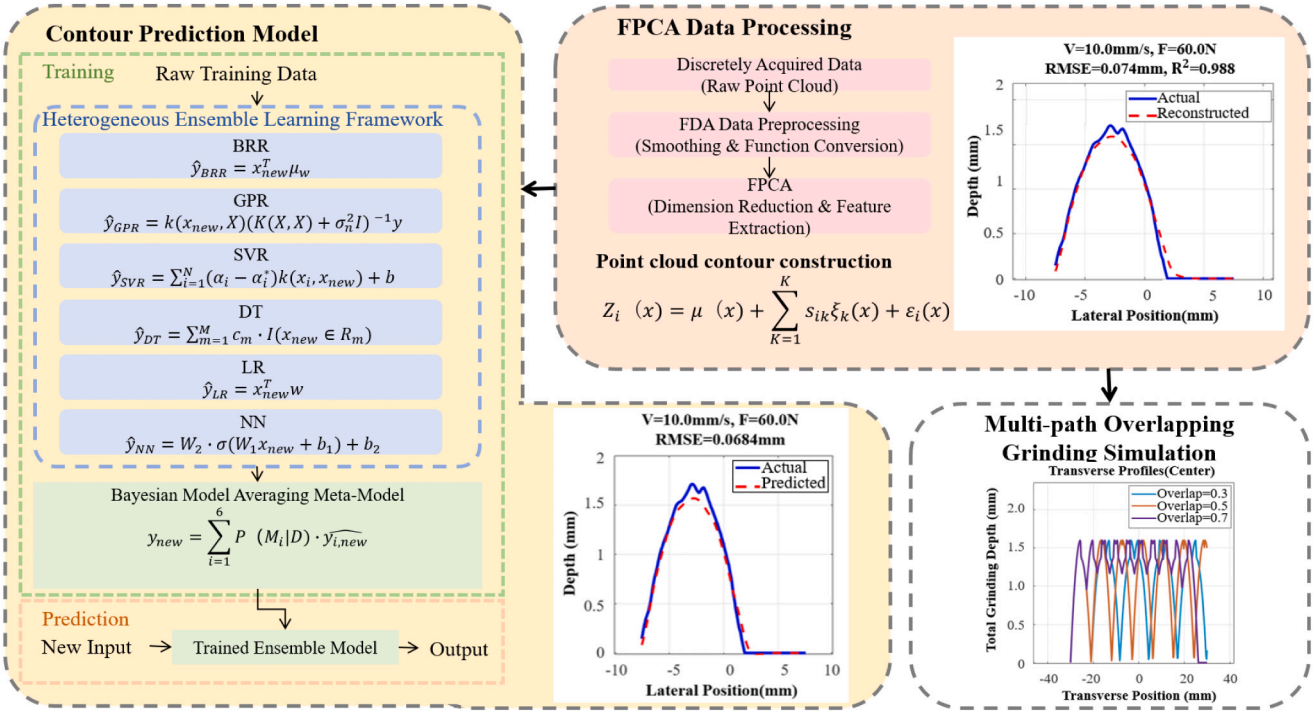


Fig. 1-2. Research framework and main content.

Bayesian hyperparameter tuning, and heterogeneous model integration, the performance of the grinding contour prediction model is improved to RMSE = 0.2205 ± 0.0228 mm, R^2 = 0.8420 ± 0.0300, and MAE = 0.1688 ± 0.0206 mm. Finally, the single-track contour prediction model is applied to simulate multi-track grinding under varying overlap rates. Simulation results demonstrate that in robotic grinding operations, the overlap rate between adjacent paths must be dynamically adjusted according to real-time contour morphology.

2. Methodology

2.1. FPCA-based reconstruction of grinding contour point cloud curves

To predict the complete grinding cross-sectional contour rather than merely geometric features such as grinding depth and width, this study employs the Functional Data Analysis (FDA) workflow illustrated in Fig. 2-1 for processing discrete point cloud data of grinding contours. This approach enables the transformation of discrete point cloud data into continuous functional representations [37]. The process begins by fitting the discrete point cloud, which may contain measurement noise,

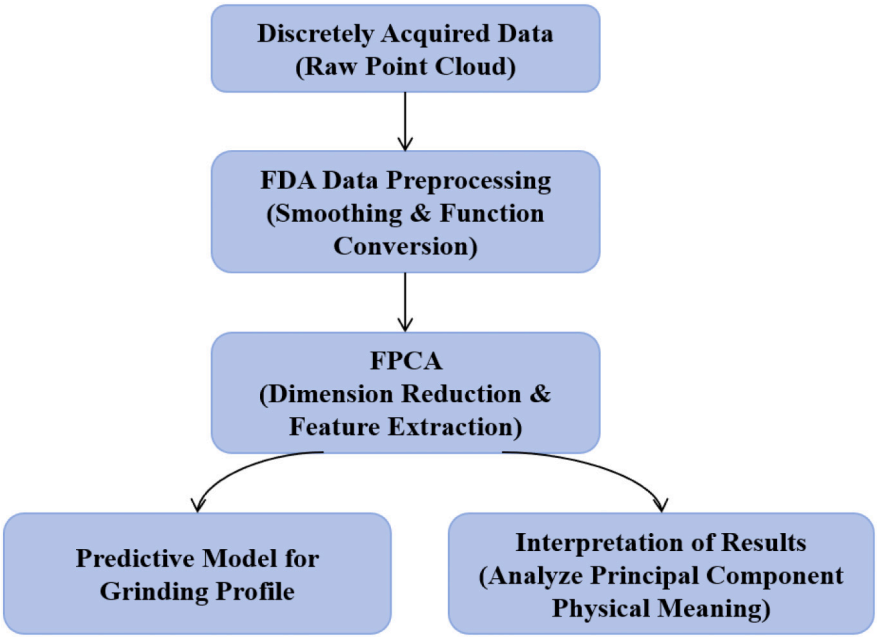


Fig. 2-1. Functional data analysis workflow for discrete grinding cross-sectional contour point cloud data.

into a smooth continuous function $X(t)$ using B-spline basis functions. Functional Principal Component Analysis (FPCA) is then applied to decompose and reduce the dimensionality of this functional representation, facilitating a physically interpretable decomposition of contour variation patterns. By establishing relationships between principal component scores and process parameters (e.g., scanning velocity V, applied force F), subsequent contour prediction models can create quantitative mappings from the process parameter space to the contour morphology space. From the perspective of model generalization and practical application, FDA processing allows the model to utilize holistic contour morphology information for prediction, thereby avoiding data loss and positional bias that may arise from discrete feature point selection at arbitrary lateral positions.

The core step is Functional Principal Component Analysis (FPCA), which aims to extract key feature patterns from high-dimensional point cloud data, achieving dimensionality reduction and feature extraction. The FPCA process is described in the following literature [38]:

First, discrete point cloud contour data is transformed into continuous functions via basis function expansion, as shown in Eq. (2-1):

$$Z_i(x) = \sum_{k=1}^{K} c_{ik}\phi_k(x) \tag{2-1}$$

where $\phi_k(x)$ denotes the basis function (e.g., B-spline basis), and c_{ik} represents the basis function coefficients. Mean function calculation as shown in Eq. (2-2):

$$\mu(x) = \frac{1}{n}\sum_{i=1}^{n} Z_i(x) \tag{2-2}$$

Eq. (2-3) shows the covariance function estimate:

$$C(x,x') = \frac{1}{n-1}\sum_{i=1}^{n} [Z_i(x) - \mu(x)][Z_i(x') - \mu(x')] \tag{2-3}$$

The principal component function is obtained by solving the Fredholm integral equation, as shown in Eq. (2-4):

$$\int C(x,x')\xi_k(x')dx' = \lambda_k\xi_k(x), k = 1,2,\ldots,K \tag{2-4}$$

where $\xi_k(x)$ denotes the k-th principal component function and λ_k represents the corresponding eigenvalue, satisfying $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_K$.

The projection score of each contour onto the principal component is given by Eq. (2-5):

$$s_{ik} = \int [Z_i(x) - \mu(x)]\xi_k(x)dx, k = 1,2,\ldots,K \tag{2-5}$$

Then any point cloud contour can be represented as a function as shown in Eq. (2-6):

$$Z_i(x) = \mu(x) + \sum_{k=1}^{K} s_{ik}\xi_k(x) + \varepsilon_i(x) \tag{2-6}$$

where $\varepsilon_i(x)$ is the reconstruction error.

The core of this representation is the set of principal component scores s_{ik} for each contour. These scores become the target variables for subsequent prediction models, effectively transforming the challenging problem of predicting an entire continuous contour shape into the more tractable task of predicting a few numerical scores from the process parameters (V,F). This foundational step enables all subsequent contour prediction and optimization.

2.2. Modeling of grinding groove cross-sectional contour

Based on functional principal component analysis, this study initially employs the conventional response surface methodology to establish predictions from process parameters to contour morphology. To address potential overfitting and prediction instability in small-sample datasets, an enhanced grinding contour prediction model integrating Bayesian optimization and ensemble learning is subsequently proposed.

2.2.1. Response surface model for grinding contour

To establish the mapping relationship between process parameters and contour shape, this study develops a quadratic response surface model for each principal component k as shown in Eq. (2-7), linking the process parameters (V,F) to the principal component scores s_k:

$$s_k = \beta_{0k} + \beta_{1k}V + \beta_{2k}F + \beta_{3k}VF + \beta_{4k}V^2 + \beta_{5k}F^2 + \epsilon_k \tag{2-7}$$

where β_{jk} represents the model coefficients and ϵ_k denotes the model error.

Based on n experimental samples, the model parameters are estimated using the least squares method as shown in Eq. (2-8):

$$\widehat{\beta} = \left(X^TX\right)^{-1}X^Ts_k \tag{2-8}$$

where the design matrix X is defined as shown in Eq. (2-9):

$$X = \begin{bmatrix} 1 & V_1 & F_1 & V_1F_1 & V_1^2 & F_1^2 \\ 1 & V_2 & F_2 & V_2F_2 & V_2^2 & F_2^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & V_n & F_n & V_nF_n & V_n^2 & F_n^2 \end{bmatrix} \tag{2-9}$$

and $s_k = [s_{1k}, s_{2k}, \ldots, s_{nk}]^T$ is the score vector of the k-th principal component.

Given new process parameters (V^*,F^*), the principal component scores are predicted as shown in Eq. (2-10):

$$\widehat{s}_k^* = \widehat{\beta} + \widehat{\beta}V^* + \widehat{\beta}F^* + \widehat{\beta}V^*F^* + \widehat{\beta}(V^*)^2 + \widehat{\beta}(F^*)^2 \tag{2-10}$$

In Eq. (2-11), the grinding contour is then reconstructed based on the predicted principal component scores:

$$\widehat{Z}^*(x) = \mu(x) + \sum_{k=1}^{K}\widehat{s}_k^*\xi_k(x) \tag{2-11}$$

The predictive performance of the model is evaluated using the metrics shown in Table 1:

2.2.2. Reverse optimization of grinding contour process parameters

The inverse problem involves defining a reverse mapping from a desired output contour to the input process parameters required to precisely generate it. This process can be formulated as a multi-objective optimization problem, where the primary goal is to achieve the target grinding depth while simultaneously ensuring the stability and feasibility of the process parameters. Given a target grinding depth D_{target}, the objective is to identify the optimal combination of process parameters (feed rate and normal force) that minimizes the absolute error between the predicted depth and the target depth, subject to the operational constraints of the robotic system. The mathematical model is formalized as Eq. (2-12):

$$\min\left|F(v,F_n) - D_{target}\right| \tag{2-12}$$

Table 1
Performance evaluation metrics for traditional response surface models of grinding contours.

Root mean square error/RMSE (mm)	Coefficient of determination/R^2	Mean absolute error/MAE (mm)
$\sqrt{\frac{1}{m}\sum_{j=1}^{m}\left[Z(x_j) - \widehat{Z}(x_j)\right]^2}$	$1 - \frac{\sum_{j=1}^{m}\left[Z(x_j) - \widehat{Z}(x_j)\right]^2}{\sum_{j=1}^{m}\left[Z(x_j) - \overline{Z}\right]^2}$	$\frac{1}{m}\sum_{j=1}^{m}\left\|Z(x_j) - \widehat{Z}(x_j)\right\|$

where $F(v, F_n)$ is the depth prediction function established by the predictive model. This optimization problem must satisfy the process constraints: $v \in [v_{\min}, v_{\max}]$ and $F_n \in [F_{n,\min}, F_{n,\max}]$.

2.2.3. Grinding contour prediction model based on Bayesian optimization and ensemble learning

Conventional response surface methodology is prone to overfitting and prediction instability under small-sample conditions. This section proposes a small-sample modeling approach integrating Bayesian optimization and ensemble learning, based on a dataset of 55 process experiments. Following the acquisition of principal component scores, a Bayesian optimization framework is introduced for automated parameter tuning [39,40]. The core concept of Bayesian optimization involves constructing a probabilistic surrogate model of the objective function to guide an intelligent search through the parameter space. Specifically, the optimization objective function [41] is defined as Eq. (2-13):

$$\mathrm{L}(\theta) = \frac{1}{N_{cv}} \sum_{i=1}^{N_{cv}} \mathrm{MSE}\left(\xi_k^{test(i)}, \widehat{\xi}_k^{test(i)}(\theta)\right) \tag{2-13}$$

where $\theta = \{\lambda, \sigma, \epsilon, \kappa\}$ represents the hyperparameter set, including key parameters such as regularization coefficients and kernel function parameters, and N_{cv} denotes the number of cross-validation folds. Bayesian optimization establishes a probabilistic model of the objective function using Gaussian processes, as shown in Eq. (2-14):

$$\mathrm{L}(\theta) \sim \mathrm{GP}(m(\theta), k(\theta, \theta^{'})) \tag{2-14}$$

where $m(\theta)$ is the mean function and $k(\theta, \theta^{'})$ is the covariance function. In Eq. (2-1), the Expected Improvement (EI) acquisition function is employed to guide the sequential search through the parameter space:

$$\mathrm{EI}(\theta) = \mathrm{E}[max(0, \mathrm{L}^{*} - \mathrm{L}(\theta))] \tag{2-15}$$

where L^{*} represents the current optimal objective value. The advantage of this method lies in its ability to rapidly converge to the optimal hyperparameter combination through a limited number of evaluations, significantly enhancing the model's baseline performance.

To address the insufficient generalization capability of single models under small-sample conditions, a heterogeneous ensemble learning framework illustrated in Fig. 2-2 is constructed. Ensemble learning is a machine learning method that combines multiple weak learners through specific strategies to form a comprehensive and robust strong learner for completing specific tasks. It has been widely adopted due to its superior accuracy, generalization capability, and robustness [42]. As illustrated in Fig. 2-2, this framework integrates six foundational learners with distinct inductive biases: Bayesian ridge regression, Gaussian process regression, support vector regression, decision tree regression, linear regression, and neural networks. Each model captures the complex mapping between process parameters and principal component scores from different perspectives: Bayesian ridge regression provides stable baseline predictions by incorporating regularized priors; Gaussian process regression quantifies prediction uncertainty based on kernel methods; support vector regression focuses on learning decision boundaries with maximum margin; decision tree regression captures local features through recursive partitioning; linear regression ensures model interpretability; neural networks learn complex function mappings via multi-layer nonlinear transformations. The synergistic operation of these heterogeneous models effectively enhances the diversity and robustness of the ensemble system.

The core of the ensemble learning framework lies in its dynamic weight allocation mechanism, which constructs a weight function based on a multi-objective consideration of both prediction accuracy and model complexity as shown in Eq. (2-16) [43]:

$$w_m = \frac{\exp(-\alpha \cdot MSE_m + \beta \cdot C_m)}{\sum_{j=1}^{M} \exp(-\alpha \cdot MSE_j + \beta \cdot C_j)} \tag{2-16}$$

where MSE_m characterizes the prediction accuracy of the m-th model, C_m

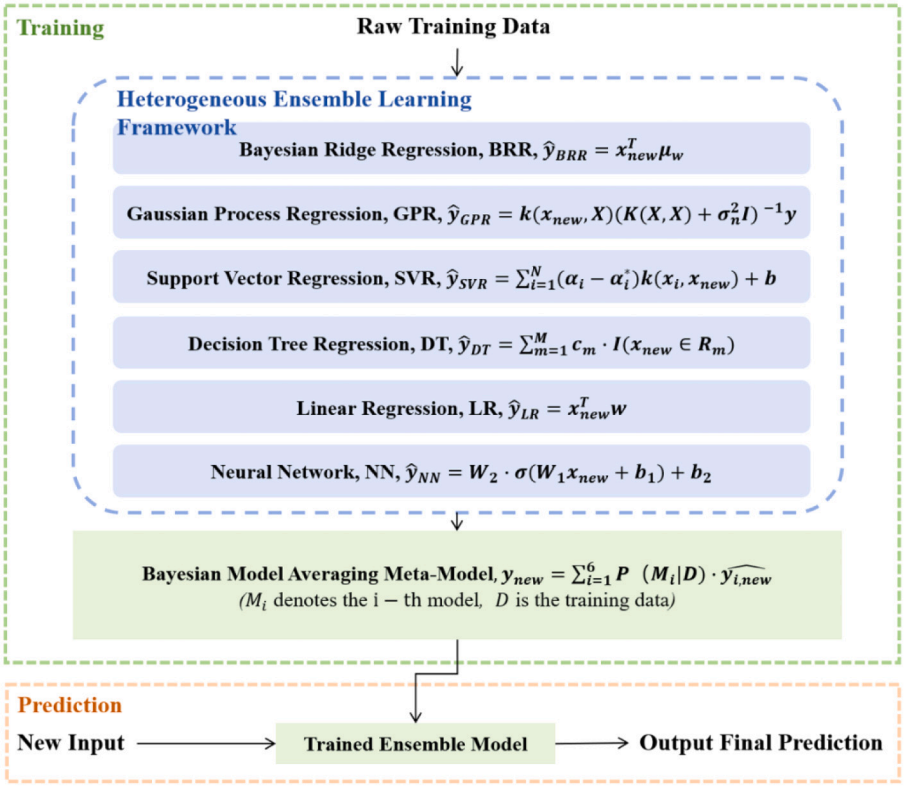


Fig. 2-2. Schematic diagram of heterogeneous ensemble learning framework.

reflects the model complexity, and α and β are balancing parameters. Through this weight allocation strategy, models with excellent performance and moderate complexity are assigned higher weights, while models with poor performance or excessive complexity are appropriately suppressed. As shown in Eq. (2-17), the final ensemble prediction is achieved through weighted combination [44]:

$$\hat{\xi}_k^{ensemble} = \sum_{m=1}^{M} w_m \cdot \hat{\xi}_k^{(m)} \tag{2-17}$$

This combination approach preserves the advantageous characteristics of each base model while mitigating the overfitting risk of individual models through weight adjustment.

3. Experiment

3.1. Experimental system

This study developed a robotic grinding process parameter acquisition and analysis platform, as illustrated in Fig. 3-1. As depicted in Fig. 3-1(b), the system features a 6-degree-of-freedom Yaskawa industrial robot equipped with a SAIWIDER ECM2A series floating module and an 8 kW electric spindle mounted at the end effector. Grinding operations were performed using a Morinaga Precision Grinding Wheel (Model E0014320726) with the following specifications: 250 mm outer diameter, 15 mm thickness, 32 mm bore diameter, and 3.5 mm chamfer radius, suitable for carbon steel processing. For precise surface topography measurement, the system incorporates a Mech-Mind Mech-Eye series 3D industrial camera that efficiently captures high-density point cloud data of ground surface contours.

The experimental workpiece consisted of Q235B low-carbon steel plates (150 mm × 100 mm × 25 mm), shown in Fig. 3-1(c). This material is widely employed in engineering machinery structures, vehicle chassis, and metal enclosures due to its excellent formability, weldability, and cost efficiency. The moderate carbon content and favorable processing characteristics of Q235B make it particularly suitable for simulating pretreatment conditions of steel components in industrial repair scenarios, where typical damage includes wear, corrosion, and fatigue cracks.

3.2. Design of process experiments

This experiment employed a full-factorial design with 55 parameter combinations (Table 2) to systematically analyze the effects of robotic feed rate (8–16 mm/s) and normal force (50–100 N) on the contour characteristics of U-shaped grinding grooves. The grinding wheel speed was held constant at 8000 rpm to minimize the influence of additional variables. The available experimental dataset was divided into training (70 %), validation (15 %), and test (15 %) subsets, with the central point condition (12 mm/s, 75 N) intentionally included in the training set.

3.3. Grinding contour point cloud data processing

To comprehensively characterize the morphological features of the grinding groove cross-section, this study developed a processing workflow for noise reduction, correction, and feature extraction of the substrate point cloud data obtained from a surface laser scanner. The procedure begins with denoising the point cloud of a representative grinding groove contour. Since the point cloud projection on the XOZ coordinate plane exhibits continuity, the data is projected onto this plane for filtering. In addition to noise, point clouds often exhibit tilting caused by uncalibrated line lasers or scanning planes during acquisition. To address this, a correction step is implemented using RANSAC-based plane fitting to determine the substrate plane, followed by adjustment of the point cloud z-coordinates relative to this reference plane. Finally, an improved K-means clustering method is employed to extract point cloud data from multiple grinding grooves.

Fig. 3-2(a) shows the multi-groove point cloud captured by a 3D camera, resulting from grinding with different process parameters. Fig. 3-2(b) presents the processed point cloud of a single groove after

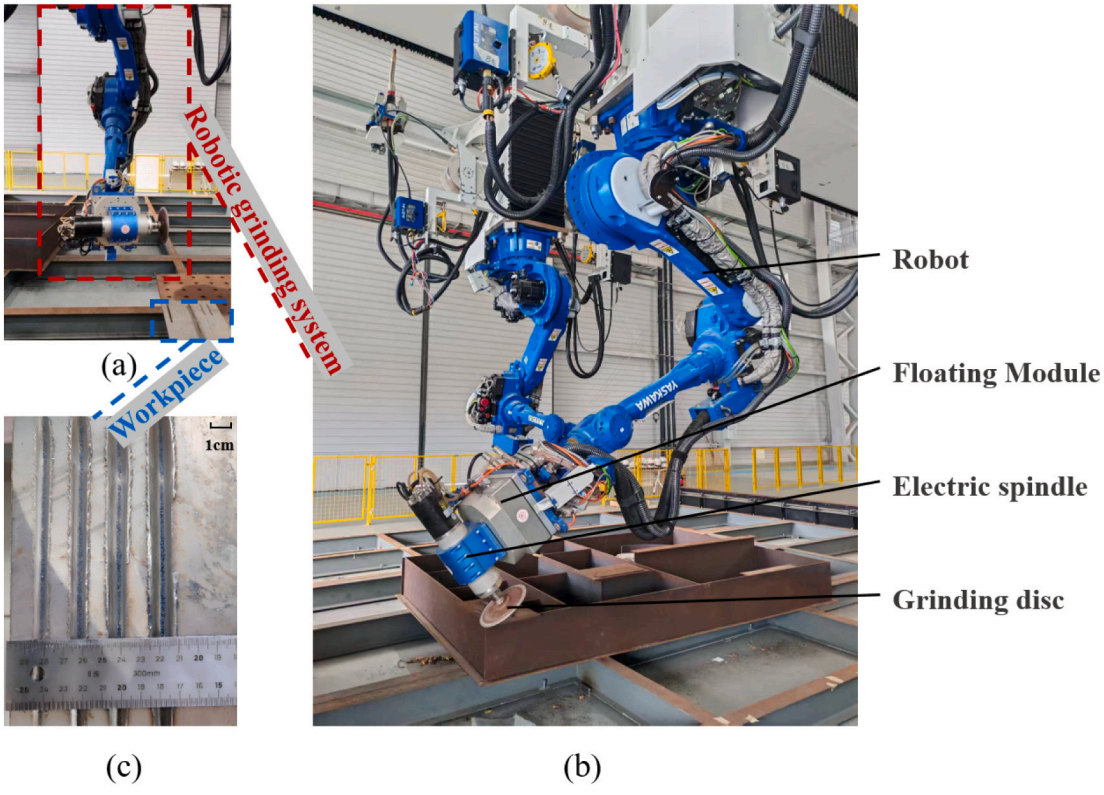


Fig. 3-1. Robotic grinding system configuration: (a) Overall robotic grinding workstation; (b) robotic grinding system; (c) the experimental workpiece.

Table 2
Summary of experimental factors and data partitioning.

No.	Feed rate (mm/s)	Normal force (N)
1	8	50
2	8	60
3	8	65
4	8	70
5	8	75
6	8	80
7	8	85
8	8	90
9	8	95
10	8	100
11	10	50
12	10	60
13	10	65
14	10	70
15	10	75
16	10	80
17	10	85
18	10	90
19	10	95
20	10	100
21	12	50
22	12	60
23	12	65
24	12	70
25	12	75
26	12	80
27	12	85
28	12	90
29	12	95
30	12	100
31	14	50
32	14	60
33	14	65
34	14	70
35	14	75
36	14	80
37	14	85
38	14	90
39	14	95
40	14	100
41	16	50
42	16	60
43	16	65
44	16	70
45	16	75
46	16	80
47	16	85
48	16	90
49	16	95
50	16	100
51	8	55
52	10	55
53	12	55
54	14	55
55	16	55

denoising, correction, and extraction. Ignoring the weld start and end morphology, the groove cross-sectional contour effectively represents the overall weld morphology. Projecting the point cloud onto the XOZ plane visualizes the grinding contour points within the coordinate system, as shown in Fig. 3-2(c), where the vertical axis corresponds to grinding depth. The groove depth contour typically shows smaller values at both ends and a larger value in the middle. The contour data acquired via 3D scanning exists as discrete point clouds height values sampled at discrete lateral positions. Essentially, this data represents the tangible manifestation of a continuous physical process. Grinding contours develop within a continuous spatial domain, possessing inherent smoothness and continuity as fundamental characteristics.

3.4. Influence of grinding wheel geometry on the cross-sectional contour

Since the grinding wheel geometry directly influences the formation of the grinding cross-sectional contour, its mechanism can be theoretically analyzed through the wheel-workpiece contact geometry model. As illustrated in Fig. 3-3, for a U-shaped grinding wheel, the theoretical contour of the grinding groove is governed by the effective abrasive distribution ring width R and the grinding depth z. The model reveals a nonlinear relationship between the grinding width w and the depth z. Initially, as z increases, the grinding width expands significantly. However, when z approaches R, the width tends toward the constant wheel width w, indicating a width saturation effect during the grinding process. This geometric constraint explains the observed saturation phenomenon in actual grinding. Consequently, this study focuses primarily on two key indicators: the grinding contour and the grinding depth.

4. Results and discussion

4.1. Functional reconstruction of grinding contour based on FPCA

To enable quantitative analysis and precise prediction of contour morphology, this study employs a functional data analysis framework where each experimental contour is treated as a continuous function. Functional principal component analysis (FPCA) is applied to reduce the dimensionality of high-dimensional contour data by decomposing it into a small set of principal component scores [45]. These principal components effectively capture core contour variation patterns, including overall depth, width, and asymmetry. The selection of the number of principal components is crucial for FPCA model performance. This study determines the optimal number using the Cumulative Percentage Variance (CPV) criterion. As shown in Fig. 4-1(a), the first five principal components explain 85.5 %, 8.6 %, 3.1 %, 1.1 %, and 0.4 % of the contour variance, respectively. Applying the commonly used statistical threshold of 95 % cumulative variance, the first three components collectively account for 97.2 % of the total variance, sufficiently capturing the primary contour variation patterns. Using too few components (e.g., two) would lead to loss of contour details such as edge sharpness, while including too many (e.g., four or more) could marginally improve accuracy but increase overfitting risks, especially under small-sample conditions.

According to the elbow criterion for component selection, Fig. 4-1(b) shows that beyond the third component, the marginal contribution of each additional component decreases significantly, and the fourth component contributes only 1.1 %. This indicates that subsequent components primarily represent noise and minor variations rather than physically meaningful shape characteristics. Moreover, using fewer components facilitates subsequent regression modeling and process parameter optimization, improving both generalizability and computational efficiency. The three-component model achieves an optimal balance between complexity and interpretability, ensuring accurate shape representation while mitigating overfitting and consistent with the principle of parsimony in statistical modeling [46].

Fig. 4-2(a), (b), and (c) further illustrates the physical significance of each principal component. The first principal component, explaining 85.5 % of the variance, primarily characterizes the overall depth of the contour. Representing the predominant variation pattern, it indicates that fluctuations in the contour's overall dimensions constitute the most significant morphological change within the studied parameter range. The second principal component, accounting for 8.6 % of the variance, governs width variations. Its functional curve displays a zero-centered contour with opposing signs on either side, capturing the “widening” or “narrowing” pattern of the contour. The third component, explaining 3.1 % of the variance, correlates with higher-order features such as asymmetry. Despite its modest contribution, it reveals potential systematic deviations in the process, possibly arising from equipment tilt,

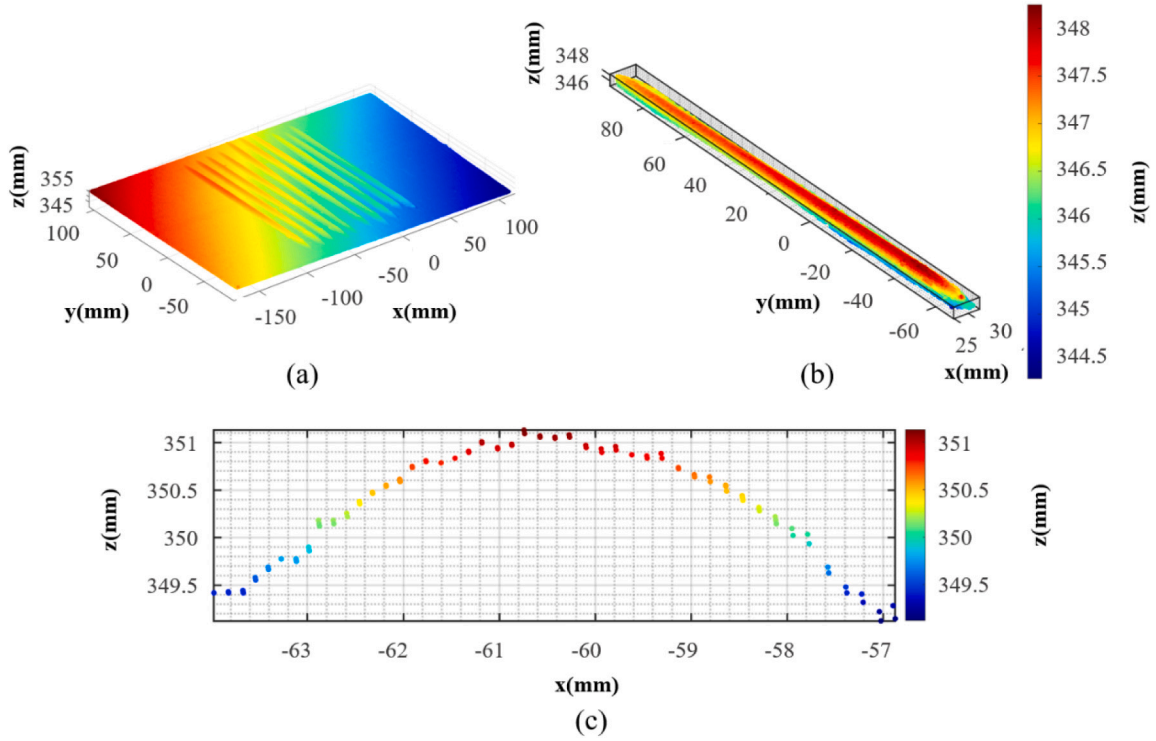


Fig. 3-2. Grinding groove point cloud projection and cross-sectional contour: (a) Original point cloud of the workpiece; (b) extracted single groove point cloud; (c) cross-sectional contour point cloud of the grinding groove on the XOZ plane.

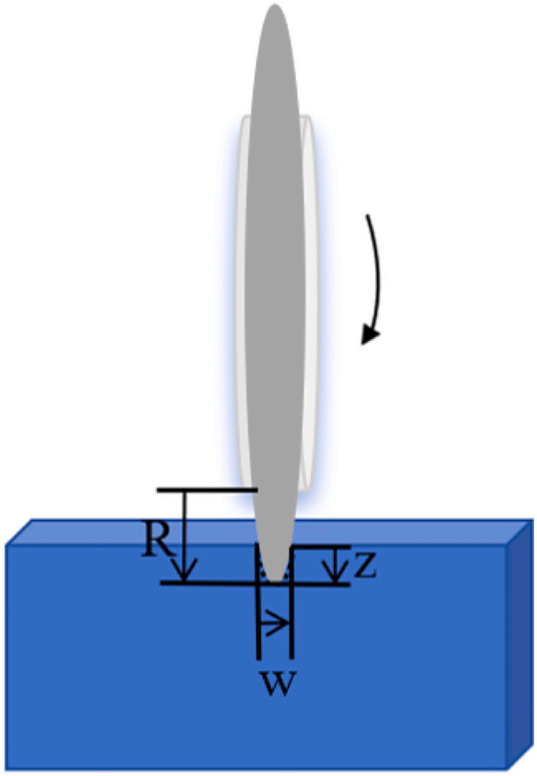


Fig. 3-3. Schematic diagram of the relationship between grinding groove contour and grinding wheel dimensions.

material heterogeneity, or uneven thermal distribution. This decomposition capability enables the model to not only accurately predict contour shapes but also provide physically meaningful insights for process optimization, offering actionable guidance for parameter adjustment.

The formation of the grinding contour is fundamentally governed by the material removal mechanism, which can be described by the Preston equation:

$$\frac{dz}{dt} = K_p \cdot P \cdot V \tag{4-1}$$

where K_p is a process constant, P is the contact pressure (proportional to the normal force F), and V is the relative feed rate. This theoretical relationship aligns with grinding dynamics, highlighting the regulatory role of process parameters on contour morphology. Experimentally, as shown in Fig. 4-3, it was observed that an increase in feed rate V with F constant leads to a wider distribution of material removal per unit time, thereby reducing the local grinding depth. Conversely, an increase in normal force F with V constant enhances the contact pressure P, directly increasing the material removal capability and grinding depth. These parameter effects are directly encoded in the FPCA results. Specifically, the first principal component-explaining 85.5 % of the contour variance and primarily characterizing the overall depth-is predominantly influenced by these same parameters. This establishes a direct and quantitative link from the process parameters to the dominant features of the reconstructed contour, providing a solid physical basis for the subsequent prediction and optimization models.

The evaluation of functional contour prediction accuracy is crucial for validating the effectiveness of the proposed FPCA method. Comparative diagrams of original versus reconstructed contours under multiple process parameters (feed rate and normal force) visually demonstrate high consistency between the reconstructed and measured contours. As shown in Fig. 4-4, the model achieves an average root mean square error (RMSE) of 0.2364 mm, indicating high precision in contour reconstruction, while the coefficient of determination (R^2) reaches 0.8826, confirming that the model captures approximately 88.26 % of contour shape variation. These results validate FPCA's capability in extracting dominant shape characteristics and establishing robust relationships between process parameters (feed rate and normal force) and

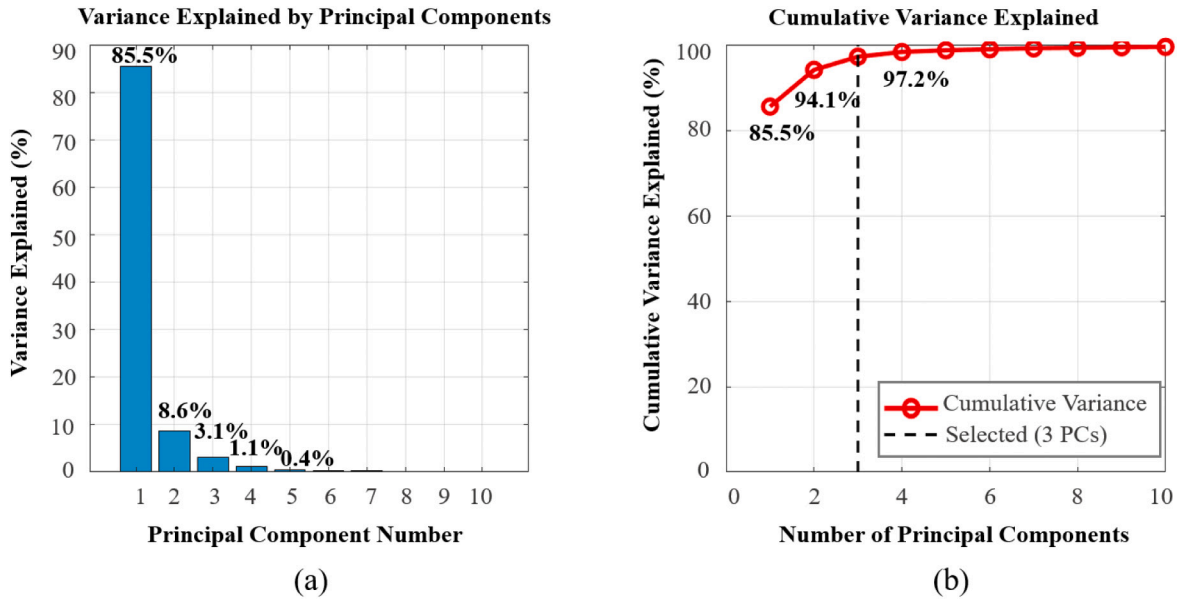


Fig. 4-1. Principal component variance explanation: (a) Variance explained by individual principal components; (b) cumulative variance contribution rate.

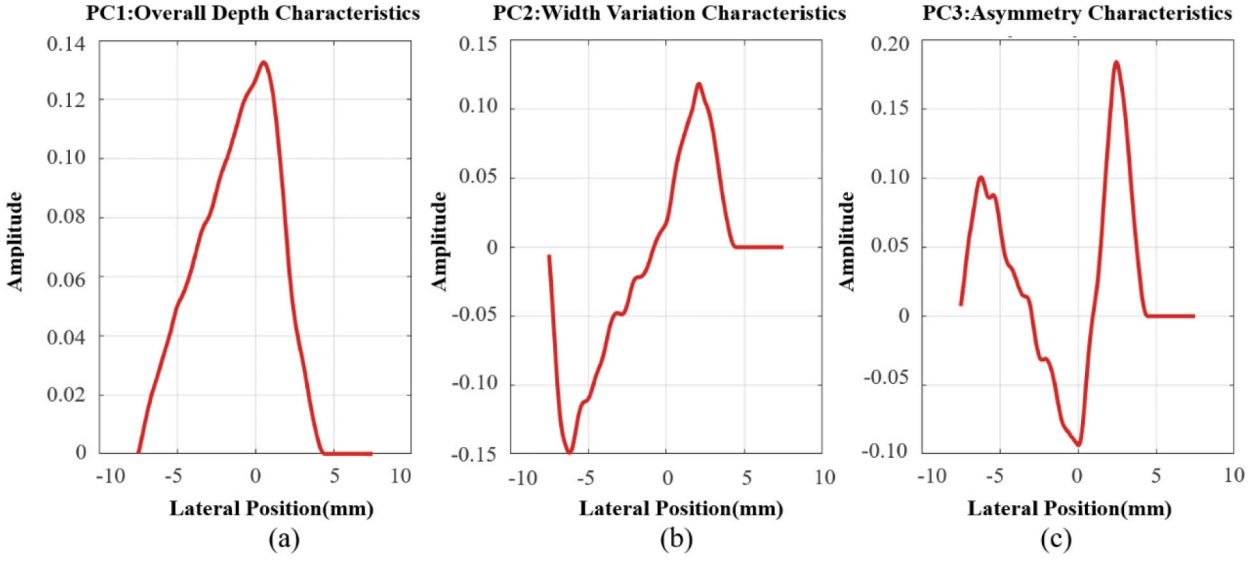


Fig. 4-2. Functional curves of principal components: (a), (b), (c) Functional curves of the first, second, and third principal components respectively.

contour features. The low reconstruction error and high explanatory power demonstrate the model's strong applicability in industrial scenarios requiring precise contour control. Building upon this foundation, subsequent phases of this study established response surface models between process parameters (V and F) and principal component scores. This dimensionality reduction strategy transforms the infinite-dimensional functional prediction problem into a manageable low-dimensional regression task, enabling complete contour morphology prediction using only V and F as inputs.

4.2. Grinding contour prediction model

4.2.1. FPCA and conventional response surface methodology-based grinding contour prediction model

Within the defined process parameter range of travel speed $V \in [8.0, 16.0]$ mm/s and normal force $F \in [50.0, 100.0]$ N, the response surface model demonstrated effective predictive capability across all experimental samples, exhibiting excellent parameter adaptability and robustness. As shown in Fig. 4-5(a) and (b), the five-fold cross-validation results indicate that the model achieved an overall root mean square error (RMSE) of 0.2591 ± 0.0176 mm, a mean absolute error (MAE) of 0.1799 ± 0.0091 mm, and a coefficient of determination (R^2) of 0.7750 ± 0.0288. These five-fold cross-validation results demonstrate that the model explains approximately 77.50 % of the contour morphology variation ($R^2 = 0.7750$), confirming its capability to capture the primary geometric characteristics. The fact that the MAE (0.1799 mm) is notably lower than the RMSE (0.2591 mm) indicates a relatively uniform distribution of prediction errors, with minimal distortion from extreme outliers. From a statistical learning perspective, the achieved R^2, coupled with low standard deviations across cross-validation folds, validates that the low-dimensional feature space constructed by the top three principal components effectively encapsulates the dominant variation patterns of grinding contours. The consistent performance across the parameter space underscores the model's robustness and reliable generalization capability. This level of predictive accuracy establishes a solid foundation for practical applications in process

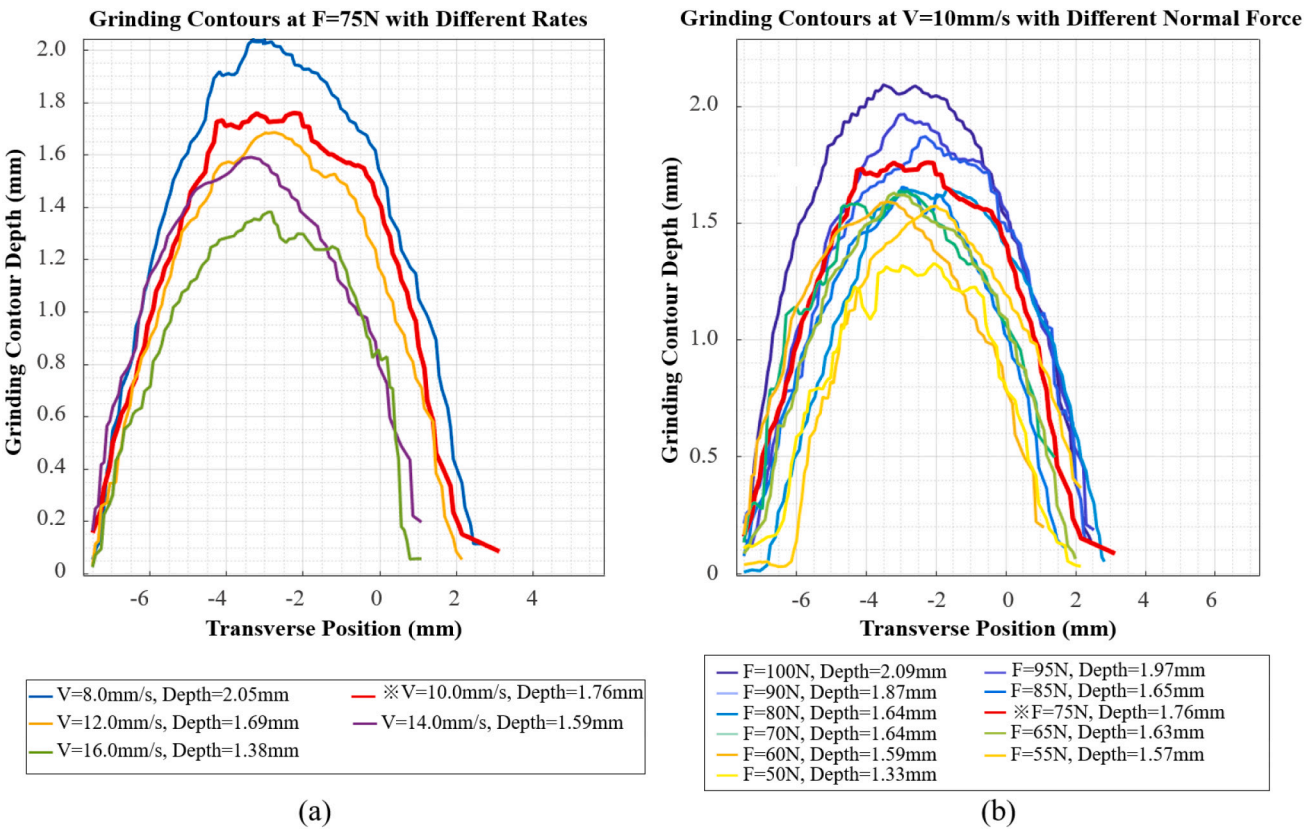


Fig. 4-3. Influence of Process Parameters on Contour Formation: (a) Grinding contours at F = 75 N with different feed rates; (b) grinding contours at V = 10 mm/s with different normal force.

parameter optimization and precise contour control.

It is necessary to emphasize the difference between the R^2 of the response surface model and the FPCA reconstruction model. The R^2 = 0.8826 of the FPCA reconstruction model (Section 4.1) quantifies the fitting degree between the denoised and smoothed reconstructed contour and the original discrete point cloud data-it is a data-driven fitting result for known contour point clouds, with the core goal of retaining inherent contour features while reducing dimensionality. In contrast, the R^2 = 0.7750 of the response surface model in this section evaluates the accuracy of mapping process parameters to contour morphology, focusing on the predictive relationship between inputs (feed rate, normal force) and outputs (contour), rather than simple fitting of existing point cloud data.

Through the established multi-objective optimization framework, the model successfully achieved reverse design of process parameters. After 39 iterations, the optimization algorithm converged to the recommended parameter combination: V = 11.87 mm/s and F = 83.61 N. As shown in Fig. 4-6, the predicted contour depth of 1.547 mm shows close agreement with the target value of 1.550 mm, validating the model's capability for precise parameter inversion. The contour prediction response surface model developed in this study, grounded in experimental data, transforms the infinite-dimensional functional prediction problem of grinding groove contours into a tractable low-dimensional regression task through FPCA-based dimensionality reduction. By establishing response surface relationships between principal component scores and process parameters (F and V), this approach maintains prediction accuracy while significantly reducing computational complexity. This methodology enables both intelligent regulation of process parameters and accurate prediction of contour contours, providing an efficient solution for grinding process optimization.

4.2.2. Grinding contour prediction model based on FPCA and Bayesian optimization

The proposed grinding contour prediction method, which integrates Bayesian optimization and ensemble learning, significantly enhances both the predictive accuracy and reliability compared to conventional response surface methodology. This improvement is achieved through FPCA-based feature extraction, Bayesian hyperparameter optimization, and the integration of heterogeneous models. Fig. 4-7 demonstrates the performance of the proposed prediction model when applied to the same process parameters previously shown in Fig. 4-5(c) from Section 4.2.1. The comparative results confirm noticeable improvement in prediction accuracy for identical contour contours. This study establishes a comprehensive optimization framework integrating FPCA, Bayesian optimization, and ensemble learning. Through strategic heterogeneous model integration and dynamic weight allocation, the framework achieves end-to-end optimization spanning from feature extraction to model construction. This integrated approach enhances model robustness and interpretability while delivering superior prediction accuracy.

Fig. 4-8 demonstrates that the optimized model achieved RMSE = 0.2205 ± 0.0228 mm, R^2 = 0.8420 ± 0.0300, and MAE = 0.1688 ± 0.0206 mm. Compared to the basic response surface model in Section 4.2.1 (RMSE = 0.2591 ± 0.0176 mm, R^2 = 0.7750 ± 0.0288, MAE = 0.1799 ± 0.0091 mm), the proposed model relatively reduced the root mean square error by 14.90 %, improved the mean absolute error by 6.17 %, and increased the coefficient of determination by 8.65 %. These quantitative metrics, particularly the smaller performance fluctuations comprehensively validate the superior prediction accuracy and robustness of the proposed methodology under small-sample conditions.

The detailed fold-by-fold performance heatmaps further reveal the internal performance structure of the two models as shown in Fig. 4-9. For the proposed model, the R^2 values for all folds are above 0.79, with three folds exceeding 0.85, demonstrating high and balanced

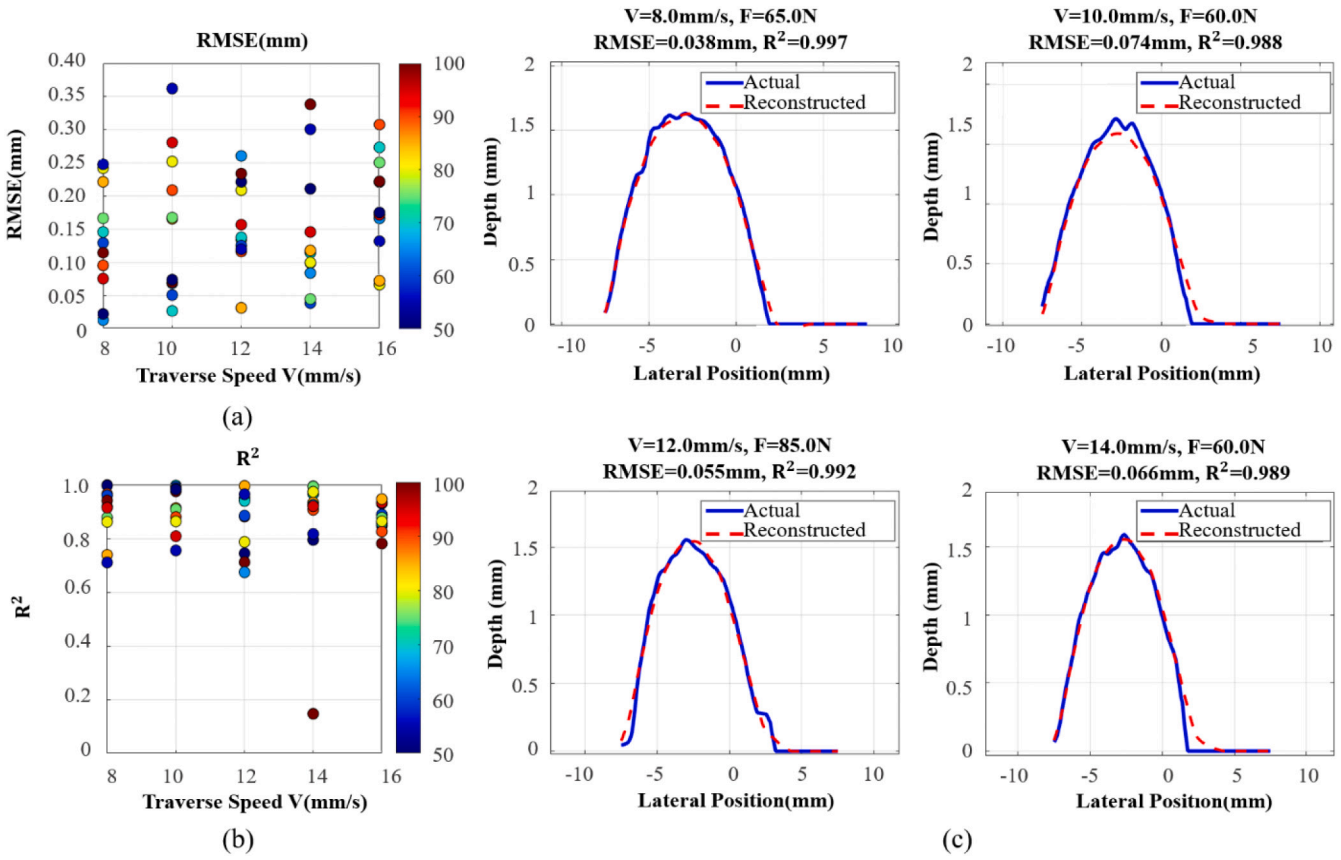


Fig. 4-4. Comparison of original contours and FPCA reconstructed contours under random process parameters and FPCA reconstruction performance: (a) RMSE distribution of FPCA; (b) R^2 distribution of FPCA; (c) comparison diagrams of original contours and FPCA reconstructed contours under different process parameters.

performance. Meanwhile, its RMSE and MAE are maintained at low and consistent levels. In contrast, the conventional RSM model shows significant performance variations across folds. For instance, the R^2 is 0.8015 for fold 3, but only 0.7727 for fold 2 and 0.7290 for fold 5. This performance instability may limit the reliability of conventional models in industrial settings. In contrast, the proposed model overcomes the challenges of small-sample learning by leveraging Bayesian optimization for automatic hyperparameter tuning and a dynamically weighted ensemble strategy to extract robust features from limited data. In summary, this work establishes an integrated framework that combines functional principal component analysis (FPCA), Bayesian optimization, and ensemble learning. The cross-validation results validate that the framework achieves end-to-end optimization-from feature extraction to model construction-enhancing not only prediction accuracy but also robustness and interpretability. This approach provides a reliable theoretical and technical foundation for precise contour control in robotic grinding processes.

4.3. *Multi-pass grinding path planning simulation*

This section simulates multi-pass grinding path planning based on the contour functions predicted by FPCA and investigates the impact of different overlap rates (0.3, 0.5, 0.7) on material removal during grinding. The overlap ratio in multi-pass grinding involves a multi-objective optimization problem, balancing surface quality, material removal rate (MRR), and defect suppression. This section simulates path planning using predicted single-pass contours and material removal physics to quantify the effects of overlap ratios (0.3, 0.5, 0.7). The overlap ratio r is expressed as:

$$r = \frac{W_{overlap}}{W_{single}} \tag{4-2}$$

where $W_{overlap}$ refers to adjacent path overlap width and W_{single} represents single-pass effective grinding width derived from single-pass contour analysis for physical rationality. Subsequently, the single-pass grinding contour is generated from the optimized model's output parameters (V, F) to reflect process parameter regulation on groove morphology, laying the foundation for path superposition and material removal calculation. Material removal follows the maximum depth principle in overlapping regions, ensuring physical consistency with the grinding process.

As shown in Fig. 4-10, the sample contour depicted in Fig. 4-10(a) yielded different material removal simulation results (Fig. 4-10(b), (c), and (d)) by applying successive overlapping rates (0.3, 0.5, 0.7). The simulation results in Fig. 4-10(e) and (f) indicate that the overlap rate, as a critical process parameter in multi-pass grinding, exerts a significant systematic influence on both material removal and surface texture morphology. Fig. 4-10(b) shows that under low overlap conditions, the surface texture exhibits distinct periodic groove features, with regular peak-valley structures forming between adjacent grinding tracks. While this morphology may be advantageous for preparing certain specialized functional surfaces, it can lead to elevated surface roughness values and pronounced stress concentration phenomena. Fig. 4-10(c) shows that under medium overlap conditions, the overlapping regions between grinding tracks form a more uniform microstructure. The height difference between peaks and valleys is significantly reduced, and the surface texture exhibits good continuity and consistency. This is conducive to achieving lower surface roughness and higher surface integrity. Fig. 4-10(d) demonstrates that high overlap machining produces highly uniform surface textures with virtually eliminated grinding

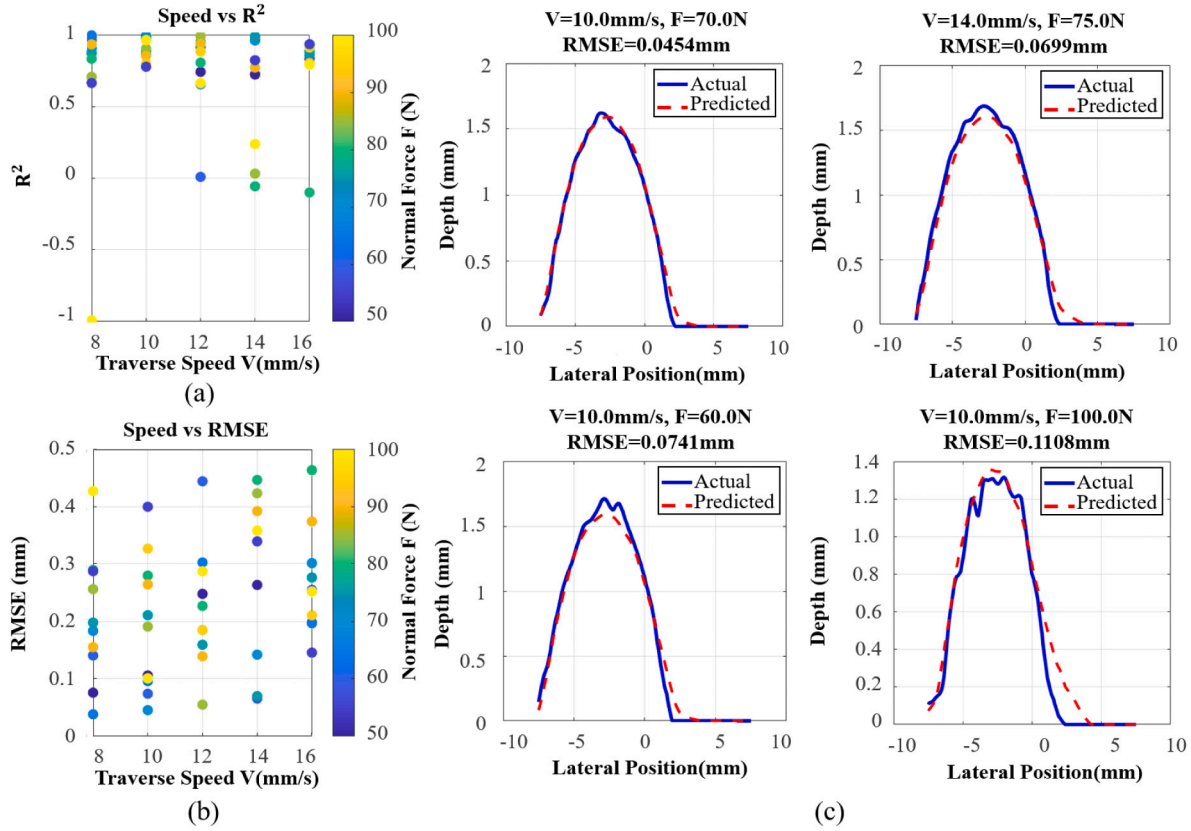


Fig. 4-5. Performance of response surface prediction model based on process experimental contour data: (a), (b) Performance metrics of the response surface prediction model; (c) comparison of original contours and response surface predicted contours under randomly selected different process parameters.

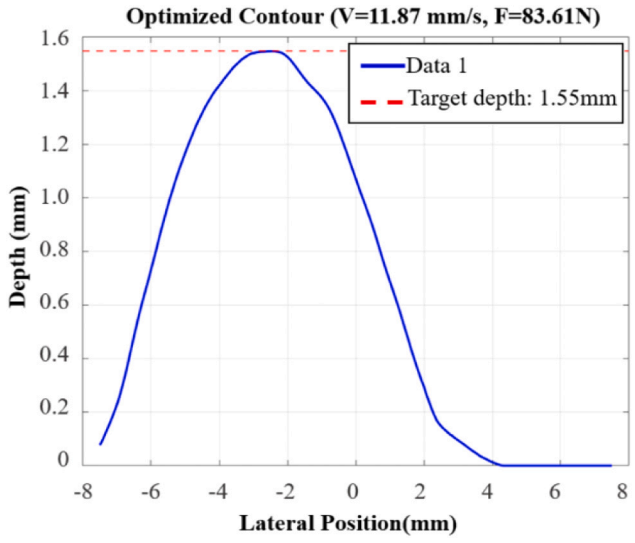


Fig. 4-6. Inverse parameter optimization of contour prediction model based on response surface methodology.

marks. However, repeated grinding operations may induce cumulative plastic deformation in the surface microstructure and generate micro-defects. Therefore, during practical process optimization, an optimal balance must be sought between material removal rate and surface texture quality based on specific machining quality requirements and efficiency targets to determine the optimal overlap rate parameters.

5. Conclusions

This paper proposes a grinding groove contour reconstruction method based on functional data analysis and establishes a predictive model for cross-sectional contours using process data. The main conclusions are as follows:

(1) A grinding groove contour reconstruction method based on functional principal component analysis (FPCA) reduces high-dimensional data to few principal components for quantitative shape analysis and prediction. The first three principal components explained 85.5 %, 8.6 %, and 3.1 % of the variance respectively, with a cumulative contribution rate of 97.2 %. And the reconstruction model achieves RMSE = 0.2364 mm and R^2 = 0.8826, which can accurately characterize the three-dimensional morphology of the grinding groove.

(2) A model combining FPCA, Bayesian optimization, and heterogeneous ensemble learning is proposed for small-sample scenarios. The ensemble framework adopts a dynamic weight allocation mechanism that balances two core objectives-model prediction accuracy and complexity to avoid overfitting while enhancing performance. The model achieves RMSE = 0.2205 ± 0.0228 mm, R^2 = 0.8420 ± 0.0300, and MAE = 0.1688 ± 0.0206 mm, enabling precise mapping from process parameters (feed rate, normal force) to contour morphology.

(3) Compared with the conventional response surface model (RMSE = 0.2591 ± 0.0176 mm, R^2 = 0.7750 ± 0.0288, MAE = 0.1799 ± 0.0091 mm), the proposed model reduces RMSE by 14.90 %, increases R^2 by 8.65 %, and decreases MAE by 6.17 %. This improvement stems from the model's inherent multi-objective trade-off between prediction precision and robustness, ensuring stronger generalization for small datasets.

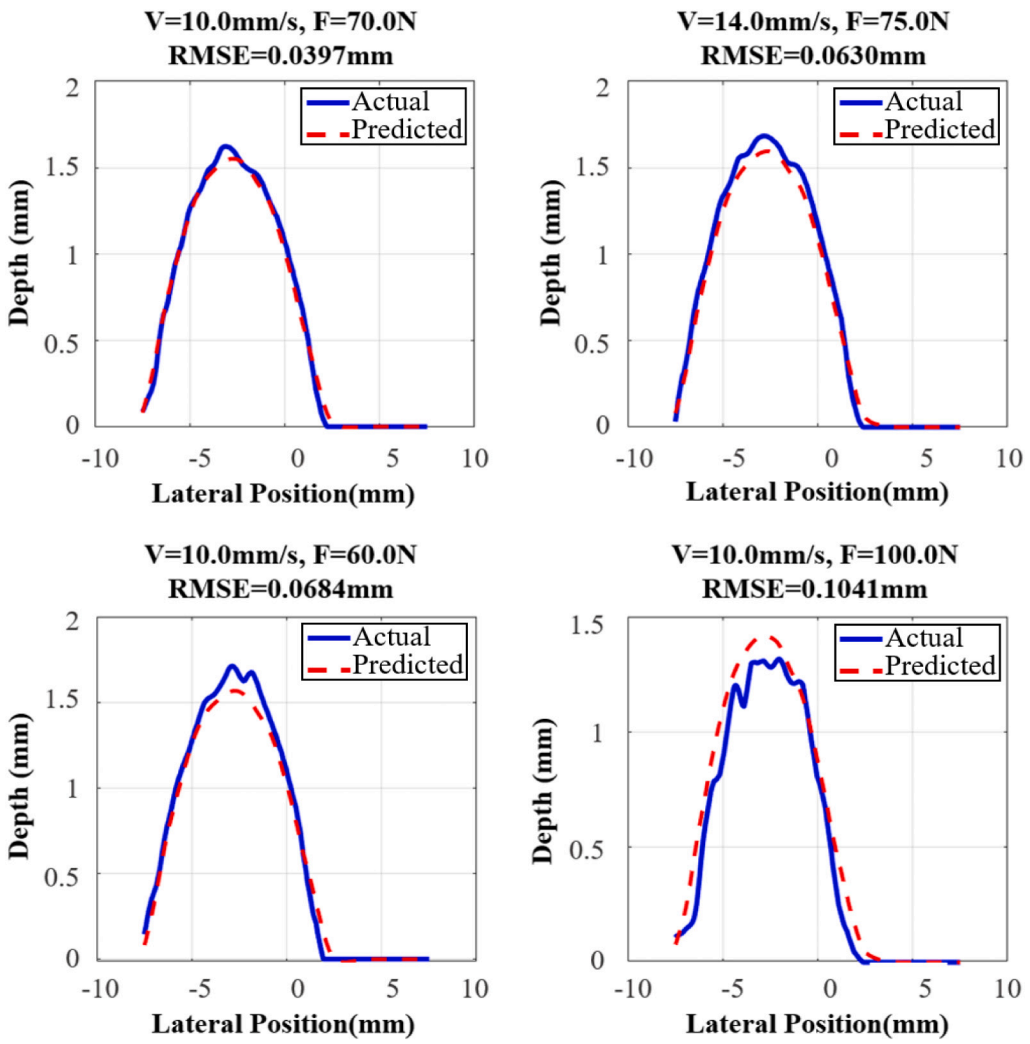


Fig. 4-7. Comparison of original contours and Bayesian optimization prediction model contours under random different process parameters.

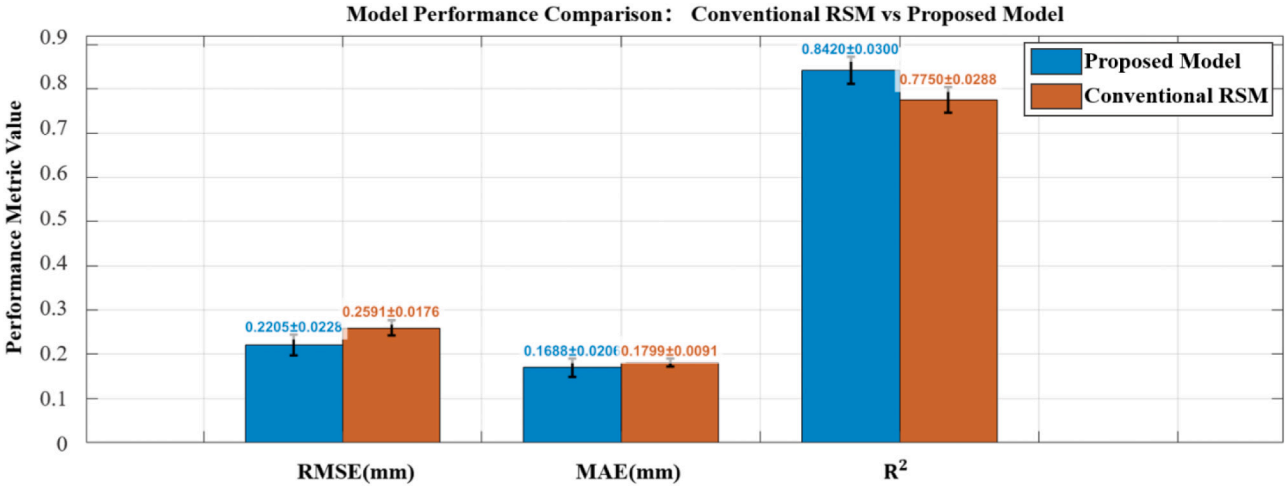


Fig. 4-8. Performance metrics of the optimized model and conventional RSM.

(4) The model addresses inverse optimization for target contour dimensions and identifies a medium overlap ratio (0.5) as optimal for balancing surface quality and material removal rate in multi-pass grinding, confirming the need for dynamic parameter adjustment in robotic path planning.

CRediT authorship contribution statement

Jingting Ji: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Jiacheng Li:** Writing – review & editing, Methodology, Data curation. **Rongqiang**

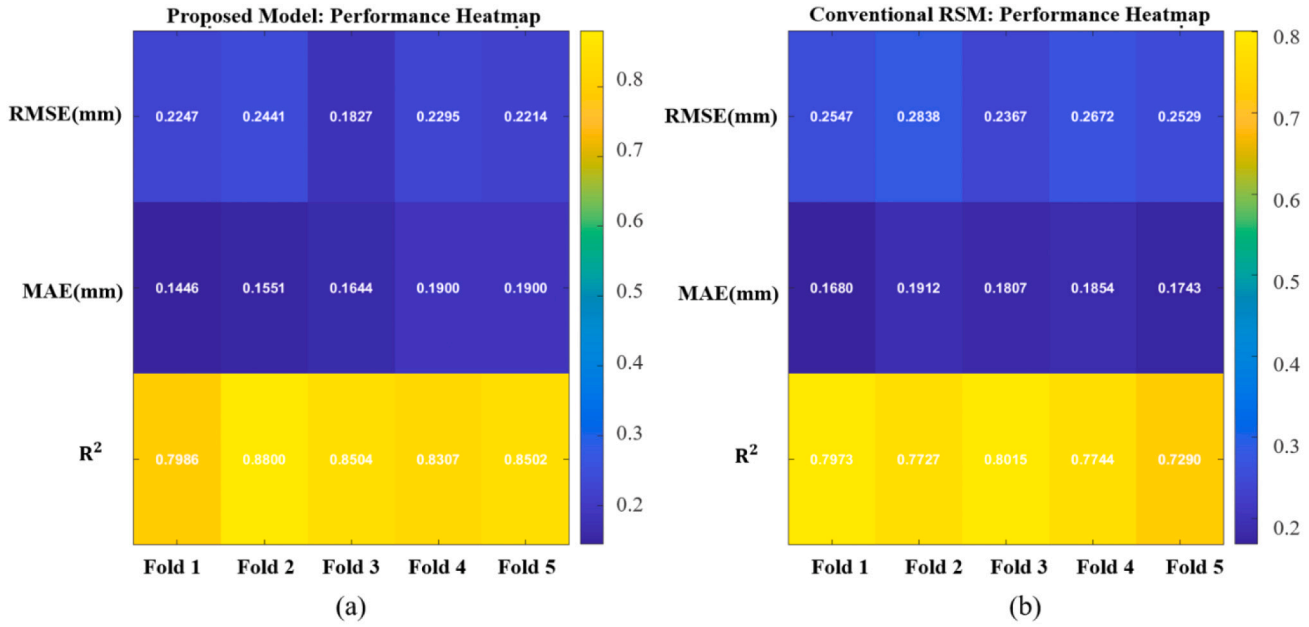


Fig. 4-9. The detailed fold-by-fold performance heatmaps of the two models: (a) The detailed fold-by-fold performance heatmaps of the proposed model; (b) the detailed fold-by-fold performance heatmaps of the conventional RSM.

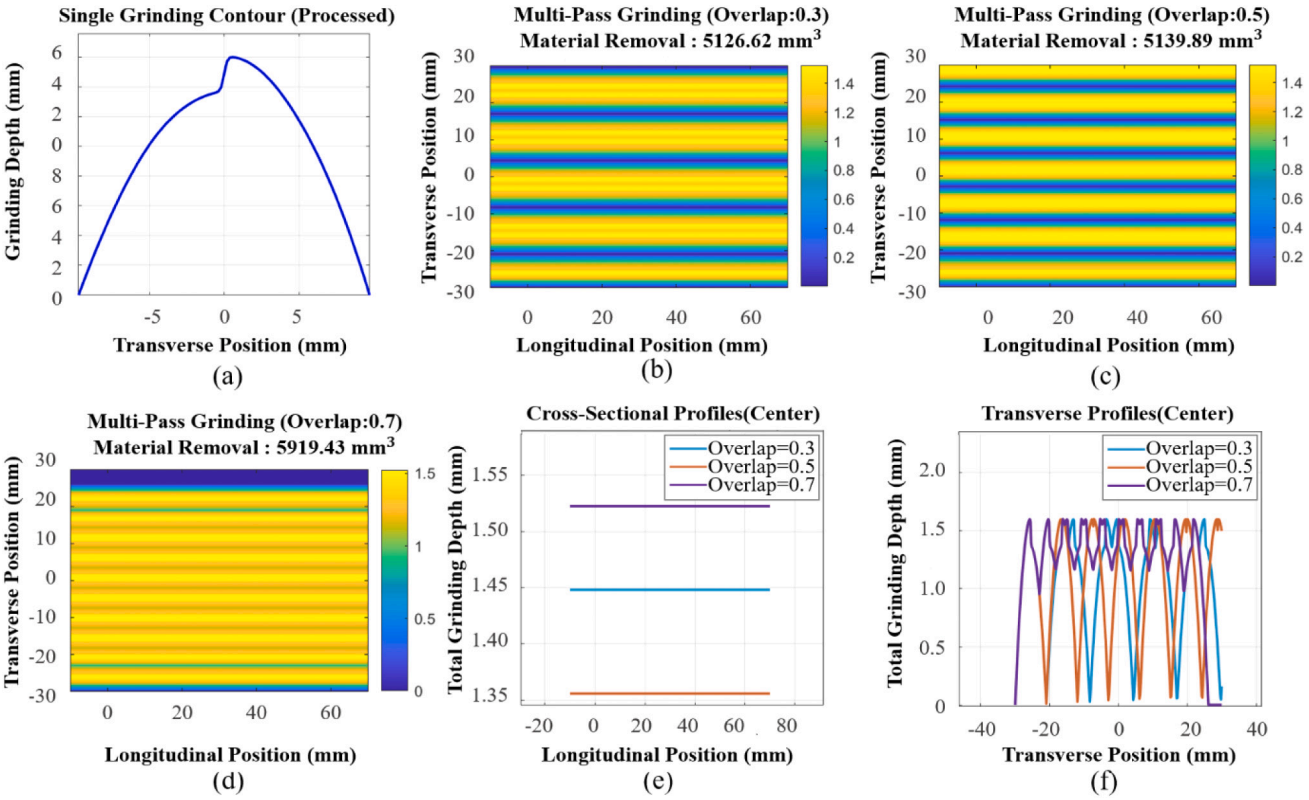


Fig. 4-10. Material removal simulation at different overlap rates (0.3, 0.5, 0.7): (a) shows a randomly selected grinding contour example; (b), (c), and (d) sequentially display multi-pass grinding removal simulations using different overlap rates (0.3, 0.5, 0.7); (e) shows the total material removal for multi-pass grinding at different overlap rates; (f) displays the contour overlap diagrams for multi-pass grinding removal at different overlap rates.

Shen: Writing – review & editing, Visualization, Data curation. **Guangchao Song:** Writing – review & editing, Data curation. **Yanling Xu:** Writing – review & editing, Supervision, Funding acquisition. **Huajun Zhang:** Writing – review & editing, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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